

Buchanan's Journal of Man, September 1887

Joseph R. Buchanan

Project Gutenberg

Buchanan's Journal of Man, September 1887

Joseph R. Buchanan

Project Gutenberg

THE PROJECT GUTENBERG EBOOK OF BUCHANAN'S JOURNAL OF MAN, SEPTEMBER 1887

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Buchanan's Journal of Man, September 1887

Editor: Joseph R. Buchanan

Release date: January 5, 2009 [eBook #27703]

Most recently updated: January 25, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/27703

Credits: Produced by Barbara Tozier, Bill Tozier and the Online
Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK BUCHANAN'S
JOURNAL OF MAN, SEPTEMBER 1887 ***

BUCHANAN'S JOURNAL OF MAN.

VOL. I.

SEPTEMBER, 1887.

NO. 8.

CONTENTS.

Concord Symposium

Rectification of Cerebral Science

Human Longevity

MISCELLANEOUS INTELLIGENCE—An important Discovery;

Jennie Collins; Greek Philosophy; Symposiums;

Literature of the Past; The Concord School; New Books;

Solar Biology; Dr. Franz Hartmann; Progress of

Chemistry; Astronomy; Geology Illustrated; A

Mathematical Prodigy; Astrology in England;

Primogeniture Abolished; Medical Intolerance and

Cunning; Negro Turning White; The Cure of

Hydrophobia; John Swinton's Paper; Women's Rights and

Progress; Co-Education; Spirit writing; Progress of the

Marvellous

Chapter VII.—Practical Utility of Anthropology (Concluded)

Chapter VIII.—The Origin and Foundation of the New
Anthropology



The Concord Symposium and their Greatest Contribution to Philosophy.

Let no one accuse the critic of irreverence, who doubts the wisdom of universities, and of pedantic scholars who burrow like moles in the mouldering remnants of antiquity, but see nothing of the glorious sky overhead. While I have no reverence for barren or wasted intellect, I have the profoundest respect for the fruitful intellect which produces valuable results—for the vast energy of the lower class of intellectual powers, which have developed our immense wealth of the physical sciences and their useful applications. Indescribably grand they are. The mathematicians, chemists, geologists, astronomers, botanists, zoologists, anatomists, and the numerous masters of dynamic sciences and arts, have lifted the world out of the ruder elements of barbarism and suffering.

But, as for the class of speculative talkers, whose self-sufficiency prompts them to assume the name of philosophers, to which they have no right, what have they ever done either to promote human welfare, or to assist human enlightenment and reveal the mysteries of life? Have they not always been as blind as owls, bats, and moles, to daylight progress? Are they not at this time utterly and *unconsciously* blind to the progress of spiritual sciences, to the revelations of psychometry and anthropology—placing themselves, indeed, in that hopeless class who are too ignorant to know their ignorance, too far in the dark to know or suspect that there is any light?

A remnant of these worshippers of antiquity still holds its seances at Concord, Mass., and publishes its amazingly dry *Journal of Speculative Philosophy*. With the unconscious solemnity of earnestness, it still digs into

Aristotle's logic and speculations—the driest material that was ever used to benumb the brains of young collegians, and teach them how *not to reason*, for Aristotle never had a glimmering conception of what the process of reasoning is. Yet all Concordians are not Aristotelians; some of them have more modern ideas, and a vigorous, though misdirected, mentality.

Prof. W. T. Harris, the leader of the Concordians, to whose lucubrations the newspapers give ample space, as those of the representative man, made a second attempt to explore the Aristotelian darkness, in which his first essay was totally lost.

If there is but a step from the sublime to the ridiculous, it is not even a step from the absurd to the ludicrous and amusing. The professional wit or joker is never so richly amusing as the man who is utterly unconscious that he is in the least funny, while heroically in earnest. The professed comedian never furnishes so much amusement as the would-be heroic tragedian, who, like the Count Joannes, furnishes uproarious merriment for the whole evening.

I have seen nothing in our Boston newspapers quite so amusing as the very friendly and sympathetic report of Prof. Harris' most elaborate and laborious comments on the SYLLOGISMS, which reminds one of Hopkinson's metaphysical and elaborate disquisition on the nature, properties, relations, and essential entity of a salt-box. We do not laugh at the professor as we did at Daniel Pratt, the "Great American Traveller," whose travels are now ended; for, aside from his metaphysical follies, Prof. Harris is a man of real merit and great intellectual industry, whose services in education will entitle him to be remembered; but when the metaphysical impulse seizes him,

"Who would not laugh if such a fool there be,
Who would not weep if Atticus were he."

The lecture of Prof. Harris was reported in the *Boston Herald*, in the style of a gushing girl with her first lover, as a "NEW STEP IN THE HISTORY OF PHILOSOPHY," attended by a full audience as "a rare treat" "*like buckwheat-cakes fresh from the griddle*," for "Prof. Harris took a decidedly *new step in*

Philosophy,” giving “an insight which *no philosopher, ancient or modern, has attained.*” Again, speaking of it privately, Prof. Harris said, “I got hold of the idea three or four years ago, and I have been trying to work it out since. I regard it as my *best contribution to philosophy.*” “*Montes parturiunt,*” What do they bring forth? Is it a mouse of respectable size? The *Boston Herald*, which is generally smart, though never profound, says of the symposium, “It has set up Aristotle this year as its golden calf to be worshipped.” “But when you ask the question, what does all this talk amount to, it is difficult to give an affirmative answer.” “It is simply threshing straw over, again and again.” But it is not aware that the Concord straw is merely the dried weeds that Lord Bacon cut up and threw out of the field of respectable literature over two hundred and sixty years ago. “What man (says the *Herald*), with any serious purpose in life, has any time to waste over what somebody thinks Aristotle ought to have thought or said.” And my readers may ask, why give the valuable space of the JOURNAL OF MAN to examining such trash? Precisely because *it is trash*, and yet occupies a place of honor, standing in the way of progress and representing the tendencies of education for centuries, which still survive, though they may be said to have gone to seed. Concord represents University philosophy, as a dude represents fashion, and as University philosophy is a haughty antagonist of all genuine philosophy, it is important to illustrate its worthlessness.

The subject of Prof. Harris’ lecture was “Aristotle’s Theory of the Syllogism, Compared with that of Hegel.” As these two were the great masters of obscurantism, the lecture should have been, of course, as perfect a specimen as either of darkness and emptiness. Omitting the definitions of syllogisms, which are familiar to all collegians, but too intolerably tedious to be inflicted on my readers, we find a very unexpected specimen of common sense following the talk about syllogisms, which embodied Aristotle’s ideas of Reason. Here it is: “Logic is often called the art of reasoning, and many people study it with a view to mastering an art of correct thinking, hoping thereby to get an instrument useful in the acquirement of truth. It may be doubted, however, whether the mind gets

much aid in the pursuit of truth by studying logic.” There is no doubt at all about it,—not one rational individual out of a hundred thousand collegians will confess that he ever got any benefit in reasoning or in pursuing truth from Aristotle’s syllogistic formula. “All men are mortal—Socrates is a man, and therefore Socrates is mortal.”

Why, then, such a flourish of trumpets over some new trick in playing with syllogism, when the whole thing is utterly worthless? And the Professor upsets himself in his own lecture, thus: “If the middle tub is contained in the big tub, and the little tub is contained in the middle tub, then the little tub is contained in the big tub.” Hegel says: “Common sense in its reaction against such logical formality and artificiality turned away in disgust, and was of the opinion that it could do without such a science as logic.” Most true, Philosopher Hegel, you have absurdities of your own on a gigantic scale, but you do well to reject the petty absurdities of Aristotle.

How does Prof. Harris rise up from Hegel’s fatal blow? He rises like Antæus from touching the earth, and triumphantly shows that syllogisms are the most necessary of all things to humanity in its mundane existence; that, in fact, we have all been syllogizing ever since we left the maternal bosom to look at the cradle, the cat, and the dog. In fact we never could have grown up to manhood, much less to be Concordian philosophers, if we had not been syllogizing all the days of our life, and, indeed, it is probable we shall continue syllogizing to all eternity, in the next life, if we have any growth in knowledge at all. Blessed be the memory of Aristotle, the great original and unrivalled discoverer of the syllogism, by means of which all human knowledge has been built up, and “blessed be the man (as Sancho Panza said) who first invented sleep,” by which we are relieved, to rest after the mighty labors of the syllogism.

And lo! we have been syllogizing all these years, alike when we listen to the nocturnal yowl of the tomcat, and to the morning song of the lark; alike, when we smell the rose, seize the orange, or devour the tempting oyster. In syllogism do we live and move, and have our being. This is the grand discovery—the last great contribution to philosophy from Concord’s greatest philosopher. We suddenly discover that we have been syllogizing

like philosophers, as Mrs. Malaprop discovered that her children had been speaking English. The illustration of this overwhelming discovery is peculiarly happy, for he applies it to the discovery of a red flannel rag in the back yard or garden, and, after detecting the red flannel by syllogism, he advances to the grander problem of showing how, by philosophic methods, we can actually distinguish an old tin can from an elephant. To enjoy this fully, the reader must take it himself from the reported lecture.

“The act of recognition is an unconscious syllogistic process in the second figure of the syllogism. I perceive something scarlet in the garden. So far I recognize a host of attributes; it is a real object; the place, surroundings and color are recognized. The sensations were so familiar that the recognition was inconceivably rapid. Then comes a slower process. The scarlet is an attribute. What can the object be? I think it is a piece of red flannel. The inference comes almost to the surface of consciousness, but I have reasoned unconsciously: This object is red. A piece of flannel is red; therefore this may be a piece of red flannel. The middle term is predicate in both premises. The unknown object is red. A familiar object (flannel) is red. Hence, I recognize this as flannel. I identify the unknown object with what is familiar in my mind. But the logician will say that this reasoning is on the invalid mode of the second figure, from which you can never draw an affirmative conclusion. Precisely so, if you mean a necessary conclusion. But sense-perception uses affirmative modes of the second figure and derives probable knowledge therefrom. I make probable knowledge more certain by verifying the inference or correcting it. I go to the garden and pick up the object, and see the threads and fiber of the wool. Or perhaps I find it was a piece of red paper. But whatever it was, at the end I can say what I have seen, only in so far as I have recognized or identified it. Recognition proceeds by the second figure, and has chiefly the non-valid modes. But it may use the valid modes, though in a still less conscious manner. For instance, I recognized that the object was not an elephant by this valid form; every elephant is

larger than a tin can; this object is not larger than a tin can; therefore, this object is necessarily not an elephant; or, by this other valid form, no elephant is as small as a tomato can; this object is just the size of a tomato can; hence this object is not an elephant. Had some one told me to look out and see an elephant, my perception would unconsciously have taken one of these forms. The scarlet is recognized as such only as it is identified with a previous impression of scarlet. Here is our third surprise in psychology. Unless there were a priori idea, sense-perception could never begin. More, unless there were a priori idea, it could not begin. For there must be two recognitions before there can be a first new idea from sense-perception. The fourth surprise is that directly with the first activity of perception in the second figure of the syllogism is joined a second activity which takes place in the form of the first figure of the syllogism. As soon as I perceived the red object to be a piece of flannel, I at once reinforced my sense-perception by unlocking all my previous store of knowledge stored up under the category of red flannel. I unconsciously syllogized thus: 'All red flannel has threads of warp and woof and a rough texture, caused by the coarse fibres of wool curling up stiffly; this is a piece of red flannel; hence this will be found to have these properties.' The act of recognition is a subsumption of the object under a class by use of the second figure of the syllogism.

"Now begins the syllogistic activity under the form of the third figure. There are a variety of attributes which I recognize by the activity of the perceiving mind in the form of the first figure, as it recognizes the general classes by the primary activity in the form of the second figure. These attributes are collected around the object as a centre of interest, and it is now the middle term. These give a new element of experience, thus: 'Major—this is a tin can; minor—it lies neglected in the garden; conclusion—tin cans get abandoned to neglect.' And so on, as to the use of the contents and the value of the can, running out into a long series of inferences."

As we have now reached the seventh heaven of Concord philosophy, and know how to distinguish an old tin can from an elephant, let us rest in peace, to meditate and enjoy its serene delights. We have had the supreme satisfaction of listening to the modern Plato, the leader at Concord. The *Herald* has informed us that on another day “the school listened with great satisfaction to Prof. Harris, who is constantly adding to the deep impression he has already made, and to the high opinion in which he is held as the most acute and profound thinker of the times, in his field.”

Lest the reader should fail to see in the foregoing what the *great contribution* to philosophy is, let us look in the *Open Court* of Chicago, which has a most affectionate partiality for metaphysical mystery. It says this “Best contribution to philosophy” “may be summed up thus,” “We can perceive nothing but what we can identify with what was familiar already.” If this were true, the babe could never perceive anything, as it begins without any knowledge, and it would be impossible for us to learn anything or acquire any new ideas. This is rather an amusing *discovery!* but it is barely possible or conceivable that there are some old fossils whose minds are in that melancholy condition.

P. S. After a few hours of repose to recover from mental fatigue and digest the new wisdom so suddenly let loose upon mankind, we discover the new aspect of the world of (Concord) philosophy. The great question of the future will be to syllogize or not to syllogize. Is it possible to distinguish an elephant from a tin can by any other method than the syllogism? When that question is decisively settled, if it ever can be settled (for metaphysical questions generally last through the centuries) Prof. Harris will have an opportunity to win still brighter laurels, and make still greater contributions to philosophy, by finding more syllogisms. Will he not prove that mathematics is the sphere of syllogism also, for if two and two make four, does not the conception of four assume the position of the major predicate, which is the generalized idea of one to a quadruple extent, and also of twos duplicated. Thus the major predicate, that four is two twos, involves the minor that two is the half of four and consequently that twice two is four. Q. E. D. The syllogism is irresistible.

If Prof. Harris should establish the mathematical syllogism and extend its power through all the realms of mathematics, as so industrious a thinker might easily do, he will have taken a step far in advance of Plato, and justly deserve a higher rank, for Plato (see his *Phædo*) was terribly puzzled over the question how one and one make two. After much puzzling he decided finally that one and one became two “by *participation in duality*.” This was the first great step to introduce philosophy into mathematics. Let Prof. Harris consummate this great work either by syllogism or by “*participation*.”

Perhaps he may introduce us to a still greater “surprise” by showing that all metaphors and poetical figures of speech are constructed on syllogistic principles. It can be done, but we must not lift the veil of wisdom too hastily, or rush in where Concord philosophers “fear to tread.” They have an endless future feast in the syllogisms, if they are faithful followers of Prof. Harris. But possibly there may be others attracted to Concord who would give the school something less dry than metaphysics, or, some other sort of metaphysics. One of their most esteemed orators made a diversion from the syllogism by presenting some other idea based on Aristotle, which ought to eclipse the syllogism, for, according to the report, he said “It is the most *momentous question that can engage the human attention*. It involves the *reality of God*, of personal existence, and freedom among men, and of immortality.”

Immense it must be! Dominic Sampson would surely say “*Prodigious!*” An attentive study of the obscure phraseology of this philosopher enables one to discover that the great and tragical question concerns the reality of reality, or what the reality is, and whether it is real or not, and how we can find it out. The way to find out whether that which we think is, is or is not, is to go back to Aristotle, who is the only man that ever understood the is-ness of the is. As the lecturer is reported to say, “The *first sign* of a movement in the right direction is the serious attention now being devoted in many quarters to the writings of Aristotle, who, in this, as in many other things, will long remain the master of those that know.” Evidently those that don’t go to Aristotle don’t know anything about life, freedom, God and

immortality. How unfortunate we are, and how fortunate the professor is, must appear by his answer to the great question, reported as follows: "Prof. Davidson discussed at length the nature of phenomena, taking the underlying basis that time and space are relations of the real to the phenomenal, and nothing but relations; also that we not only have ideas of reality, but that *these ideas are the realities themselves*. Then the question is, if the *concept of reality be reality itself*, how is this related to phenomena? There is a double relation, active and passive. * * * Eternal realities are known to us only as terms of phenomena. They are in ourselves, and from the exigencies of our intelligence."

Thus we understand nothing whatever exists but our own cogitations, or, as the sailor jocosely expressed it—"Tis all in my eye"—and after these many years we are brought back to the famous expression of the Boston Transcendentalist, "we should not say *it rains, it snows*, we should say *I rain, I snow*." This, gentle, patient reader, is no burlesque, that you have been reading, it is the wisdom of the Concord Symposium of professors and authors meeting near the end of the 19th century, and basking in the smiles of *cultured* Boston! or at least that portion which is devoted to the Bostonese idea of philosophy, and thinks the feeblest glimmer of antiquity worth more than the science of to-day. Such indeed are the sentiments of the President of Boston University. And as for the wisdom of Concord, the *Open Court*, which is good authority, says: "Dr. Harris and Prof. Davidson are, without doubt, the *pillars of the school*; but there is some difference of opinion as to which is its *indispensable support*." An intelligent spectator would say that more metaphysical acumen and vigor has been displayed by DR. EDWARD MONTGOMERY than by all the remainder of those engaged in the blind hunt for philosophy at Concord.

On the last day of the Symposium, July 28, the report says "The burden has fallen wholly upon Prof. Harris, and he has borne it so as to excite the *wonder and admiration* of his listeners. He *went to the very bottom of things* as far as human thought could go, and there, as he put it, was on solid rock, with no possibility of scepticism. Both his forenoon and evening lectures were *masterly in their way*." Exactly so; they were unsurpassed as a

reproduction of the style and manner of the Aristotelian folly which held Europe fast in that wretched period called the Dark Ages, which preceded the dawn of intelligence with Galileo.

About one half of the reported lectures on Aristotle is, though cloudy, intelligible. The remainder is a fair specimen of that skimmy-dashy style of thought which glances over the surfaces of things and never reaches their substance or reality, yet boasts of its unlimited profundity because it does not know the meaning of profound. Such thinking must necessarily end in falsity and folly, of which the lecture gives many specimens, which it is worth while to quote, to show what the devotees of antiquity call philosophy—thus:

“If we cannot know the ultimate nature of being, then philosophy is impossible, for philosophy differs from other kind of knowing by seeking a first principle.” “The objects of philosophy then include those of ontology. They are first the nature of the ultimate being of the universe, the first principle, the idea of God.”

This is not philosophy, but might be called theology, and not legitimate theology even, but supra-theological—for all sane theology admits that man cannot know God. It is a desperate, insane suggestion that we must know the unknowable, and that if we cannot do that we can have no philosophy. Of course men who think this way know nothing of philosophy, and are beyond the reach of reason.

Again, “in the nature of the truly independent and true being, it sees necessary transcendence of space and time, and this is essential immortality.” This is a fair specimen of the skimmy-dashy style. Immortality is not a “transcendence of space,” if that means anything at all, but a conscious existence without end. Perhaps by “transcendence of space” he means filling all the space there is, and going considerably beyond it where there is no space.

His idea of infinity is worthy of Aristotle or Hegel, to whom, in fact, it belongs—he says, “self-conditioning is the form of the whole, the form of

that *which is its own other*.” That something should be “its own other” is just as clear as that it should be its own mother or father. Do such expressions represent any ideas, or do metaphysicians use words as a substitute for ideas—verily they do, in Hegelian metaphysics, and the same thing is done in asylums for the insane.

Again, “our knowledge of quantity is a knowledge of what is universal and necessary, and *hence* is not derived from experience.” If this is true of the professor, he knew all of mathematics before he opened his eyes in the cradle. Common mortals know nothing of quantity or anything else, until they have had a little experience. If we know everything that is “universal and necessary” without experience, the little babes must be very wise indeed.

Again, “causal energy is essentially a *self-separation*, for in order that a cause A. may produce an effect in B. outside of it, cause A. must detach or separate from itself the influence or energy which modifies B.” What does the earth *detach from itself* when it causes a heavy body to fall? In chemical catalysis what does the second body “detach from itself” to produce change in the first, which is changed by its mere presence. The assertion is but partially true, applying only to the transfer of force when one body strikes another. Aristotle has some thoroughly absurd suggestions on the same subject which Professor H. did not reproduce.

How does he grapple with the idea of God, which is the essence of his philosophy? Here it is: “The first principle as pure self-activity, must necessarily have the permanent form of *knowing of knowing*, for this root form of self-consciousness is entirely self-related. The self sees the essential self, the self-activity is the object of self.” We are instructed! God *knows he knows*, and that is the very essence of his divinity—that is enough. In this profound expression we have the consummation of philosophy, for the purpose of his philosophy is to know God, “*Nunc dimittis*,” we need to know nothing more,—*we know we know*, and so we are God’s. “This line of thought brought up at every step some phase of Plato and Aristotle,” said the professor, and we are thankful that he did not resurrect any more of the puerilities of Athenian ignorance. “Knowing of knowing” is quite enough,

which he repeats to be emphatic. “All true being is in the form of the infinite or self-related, and related to itself as the *knowing of knowing*. All beings that are not this perfect form of self-knowing, either potentially or actually, must be parts of a system or world order which is produced in some way by true being or self-knowing. All potential self-knowings contain within themselves the *power to realize* their self-knowledge, and are therefore free beings.” This is a broad hint that men are gods and lands us in that realm of folly of which Mrs. Eddy is the presiding genius. She is much indebted to the Concord philosophers for lending their respectability to her labyrinth of self-contradictions.

One quotation more, to give the essence of this Concord philosophy. “The Divine Being exists for himself as one object. This gives us the Logos, or the only-begotten. The Logos *knows himself* as personal perfection, and also as *generated*, though in an infinite past time. This is its recognition of its first principle and its unbegotten ‘Father.’ But whatever it knows in self-consciousness, it creates or makes to exist,” and more of the same sort.

We are overwhelmed with such a flood of wisdom! How the professor attained so intimate, familiar, and perfect a knowledge of the infinite power, to which the fathomless depths of starry infinity are as nothing, is a great mystery. Was it by *Kabbala* or by *Thaumaturgy*, or did he follow the sublime instructions of his great brother Plato, and thrust his head through the revolving dome of the universe, where the infinite truth is seen in materialized forms.

The “Divine” Plato (of whom Emerson said, “Plato is philosophy, and philosophy is Plato”) described the immortal Gods as driving up in chariots through the dome of the heavens to *get upon the roof*, and look abroad at infinite truth, as they stand or drive upon the revolving dome, followed by *ambitious souls who barely get their heads through the roof* with difficulty, and catch a hasty glimpse of infinite truth, before they tumble back, or lame their wings, or perhaps drop into the body of some brute. The revolving dome and the ambitious souls peeping through the roof, would be a good subject for the next symposium. They might tell us whether these ambitious

souls that peep through the roof are Concordian philosophers, or belong to the schools of Aquinas and *Duns Scotus*.

The philosophy of the Greeks is worth no more to-day than their chemistry or their physiology. The lingering superstition of believing because they had famous warriors, orators, statesmen, historians, poets, and sculptors, while entirely ignorant of science and philosophy, that their philosophic puerilities are worthy of adoration in the 19th century, a superstition which makes a fetish of the writings of Plato and Aristotle, has been tolerated long enough, and as no one has attempted to give a critical estimate of this effete literature since Lord Bacon did something in that way, I shall not much longer postpone this duty.

RECTIFICATION OF CEREBRAL SCIENCE.—In the October number the rectification of cerebral science as to psychic functions will be shown by appropriate engravings, showing how far the discoveries and doctrines of Gall and Spurzheim are sustained by positive science. In the further development of the subject, hereafter, the true value and proper position of the discoveries of Ferrier, and the continental vivisectionists will be explained, though but meagre contributions to psychology, they furnish very valuable additional information as to the functions of the brain.

Human Longevity.

Is not longevity in some sense a measure of true civilization or improvement of the race? It is certainly an evidence of conformity to the Divine laws of life and health, which reward right action with happiness, health, and long life. I cannot, therefore, think the study of longevity unimportant. To every one of us it is a vital question, for death is regarded as the greatest calamity, and is the severest penalty of angry enemies, or of outraged laws.

It is our duty as well as privilege to perfect our constitution, and see that it does not wear out too soon, that we are not prematurely called away from our duties. And I bring it as serious charge against modern systems of education, that they tend to degenerate mankind, to impair the constitution and to shorten life. That we should not submit to this, but should all aspire to live a century or longer, if we have a fair opportunity, I seriously maintain, and that my readers may be inspired with a like determination, I take pleasure in quoting examples.

In Dr. Cohausen's *HERMIPPUS REDIVIVUS* republished in 1744, I find the following statements: "It is very remarkable, that not only the sacred writers, but all the ancient Chaldean, Egyptian, and Chinese authors speak of the great ages of such as lived in early times, and this with such confidence that Xenophon, Pliny, and other judicious persons receive their testimony without scruple. But to come down to later times, Attila, King of the Huns, who reigned in the fifth century, lived to 124, and then died of excess, the first night of his second nuptials with one of the most beautiful princesses of that age. Piastus, King of Poland, who from the rank of a peasant was raised to that of a prince, in the year 824, lived to be 120, and

governed his subjects with such ability to the very last, that his name is still in the highest veneration amongst his countrymen. Marcus Valerius Corvinus, a Roman Consul, was celebrated as a true patriot and a most excellent person in private life, by the elder Cato, and yet Corvinus was then upwards of a hundred. Hippocrates, the best of physicians lived to an 104, but Asclepiades, a Persian physician, reached 150. Galen lived in undisturbed health to 104; Sophocles, the tragic poet, lived to 130; Democritus, the philosopher, lived to 104; and Euphranor taught his scholars at upward of 100; and yet what are these to Epiminedes of Crete, who, according to Theopompus, an unblemished historian, lived to upwards of 157. I mention these, because, if there be any truth or security in history, we may rely as firmly on the facts recorded of them as on any facts whatever. Pliny gives an account that in the city of Parma, there were two of 130 years of age, three of 120, at a certain taxation, or rather visitation, and in many cities of Italy, people much older, particularly at Ariminum, one Marcus Apponius, who was 150. Vincent Coquelin, a clergyman, died at Paris in 1664, at 112. Lawrence Hutland, lived in the Orkneys to 170. James Sands, an Englishman, towards the latter end of the last century, died at 140, and his wife at 120. In Sweden, it is a common thing to meet with people above 100, and Rudbekius affirms from bills of mortality signed by his brother, who was a bishop, that in the small extent of twelve parishes, there died in the space of thirty-seven years, 232 men, between 100 and 140 years of age, which is the more credible, since in the diet assembled by the late Queen of Sweden, in 1713, the oldest and best speaker among the deputies from the order of Peasants was considerably above 100. These accounts, however, are far short of what might be produced from Africa and North America, that I confine myself to such accounts as are truly authentic.” All of these instances the doctor sustains by reference to his authorities.

To the foregoing he adds the examples of teachers and persons who associate with the young, to which he ascribes great value in promoting longevity. Thus, “Gorgias, the master of Isocrates, and many other eminent persons, lived to be 108. His scholar, Isocrates, in the 94th year of his age

published a book, and survived the publication four years, in all which time he betrayed not the least failure, either in memory or in judgment; he died with the reputation of being the most eloquent man in Greece. Xenophilus, an eminent Pythagorean philosopher, taught a numerous train of students till he arrived at the age of 105, and even then enjoyed a very perfect health, and left this world before his abilities left him. Platerus tells us that his grandfather, who exercised the office of a preceptor to some young nobleman, married a woman of thirty when he was in the 100th year of his age. His son by this marriage did not stay like his father, but took him a wife when he was twenty; the old man was in full health and spirits at the wedding, and lived six years afterward. Francis Secordo Horigi, usually distinguished by the name of Huppazoli, was consul for the State of Venice in the island of Scio, where he died in the beginning of 1702, when he was very near 115. He married in Scio when he was young, and being much addicted to the fair sex, he had in all five wives, and fifteen or twenty concubines, all of them young, beautiful women, by whom he had forty-nine sons and daughters, whom he educated with the utmost tenderness, and was constantly with them, as much as his business would permit. He was never sick. His sight, hearing, memory, and activity were amazing. He walked every day about eight miles; his hair, which was long and graceful, became white by the time that he was four-score, but turned black at 100, as did his eyebrows and beard at 112. At 110 he lost all his teeth, but the year before he died he cut two large ones with great pain. His food was generally a few spoonfuls of broth, after which he ate some little thing roasted; his breakfast and supper, bread and fruit; his constant drink, distilled water, without any addition of wine or other strong liquor to the very last. He was a man of strict honor, of great abilities, of a free, pleasant, and sprightly temper, as we are told by many travellers, who were all struck with the good sense and good humor of this polite old man.”

“In the same country (as Thomas Parr) lived the famous Countess of Desmond. From deeds, settlements, and other indisputable testimonies it appeared clearly that she was upwards of 140, according to the computation

of the great Lord Bacon, who knew her personally, and remarks this particularity about her, that she thrice changed her teeth.”

The stern scepticism of the medical profession and especially among its leaders has borne so heavily against all cheerful views of life and longevity, that at the risk of becoming monotonous I again refer to this subject and present examples of longevity which cannot be denied, in addition to the list previously given. Medical collegiate scepticism can deny anything. Ultra sceptics deny centenarian life, as they also denied the existence of hydrophobia, while those who admitted its existence denied its curability.

Connecticut alone furnishes a good supply of centenarians. Three years ago Mr. Frederick Nash, of Westport, Conn, published a pamphlet giving the old people living in Connecticut, including twenty-three centenarians, whom he described. The names of twelve of these were as follows:

Edmund R. Kidder, of Berlin, Aug. 17, 1784.
Jeremiah Austin, Coventry, Feb. 10, 1783.
Mrs. Lucy Luther, Hadlyme, Jan. 6, 1784.
Walter Pease, Enfield, March 29, 1784.
Egbert Cowles, Farmington, April 4, 1785.
Mrs. Eunice Hollister, Glastonbury, Aug. 9, 1784.
Mrs. Elsie Chittenden, Guilford, April 24, 1784.
Miss Eunice Saxton, Colchester, Sept. 6, 1784.
Marvin Smith, Montville, Nov. 18, 1784.
Mrs. Phebe Briggs, Sherman, Nov. 16, 1784.
Mrs. Elizabeth Buck, Wethersfield, Jan. 10, 1784.
Mrs. Clarissa D. Raymond, Milton, April 22, 1782.

The others are either of foreign birth or former slaves, whose precise ages cannot be established.

In addition to this list the newspapers gave us Mrs. Abigail Ford of Washington, born in 1780, Mr. Darby Green of Reading, born in 1779, Tryphena Jackson, colored, born in 1782, and Wm. Hamilton, Irish, also in 1782; and an old sailor in New Haven town house claims to have been born in 1778.

The very careful investigation of Connecticut by Mr. Nash shows that "the duration of human life in this State is greater than it was a generation ago. Then only one person in 500 lived to see 80 years. Now one per cent of the population live to that age. The average age of 6,223 persons is 83 years. The number of ages ranging from 84 to 89 years is large, and those who are 90 and over number 651; nine are 99, thirteen are 98, and eleven are 97. No age of less than 80 years has been recorded.

"It may be pleasing to our grandmothers to know that in this list of more than 6,000, more than 4,000 are women, and that only eight of the twenty centenarians are men. The list adds strength to what has already been held as true, that married people always live longer than single, and it also shows that two spinsters have begun their second century. They are accompanied on the list by two sturdy bachelors."

In a sketch of centenarians published in November, 1884, are given the names of Nathaniel H. Cole of Greenwich, R. I., born in 1783, Royal C. Jameson, Papakating, N. J., born in 1784, Wm. Jovel of New Jersey, and Luther Catlin of Bridgewater, Pa., born in 1784. The last three took an active part in the last presidential election.

In Maine were reported Mrs. Sally Powers, Augusta, believed to be born in 1778, Mrs. Thankful Donnel of West Bath, 101, Mrs. Betsy Moody, 102, Mrs. Philip Pervear of Sedgwick, 105, Jotham Johnson of Durham, 100, Mrs. Small of Bowdoinham, 100. If alive to-day, they are three years older.

In Vermont, from 1881 to 1884, sixteen centenarians died; and in the last census of the United States there were 322.

In looking over my records I find so many other examples of centenarian life that I shall not weary the reader by their repetition, but examples running for over a century may be worth mentioning. Madame Lacene, one of the most brilliant women of France, died a few years ago at Lyons in her 104th year. Her will was under contest on account of her extreme age, but the court was fully satisfied of her intellectual competence. In the olden time she had often entertained Mme. de Stael, Mme. Recamier, and Benj. Constant.

The oldest person in France, perhaps in the world, is said to be a woman who lives in the village of Auberive, in Royans. She was born March 16, 1761, and is therefore 125 years old. The authentic record of her birth is to be found in the parish register of St. Just de Claix, in the department of the Isere.—*Scientific American*.

“Among the professors at German universities there were no fewer than 157 between the ages of seventy and ninety, of whom 122 still deliver lectures, seven of these being between eighty-five and eighty-nine years of age. The oldest, Von Ranke, was in active service in his 90th year. Elennich, of Breslau, only thirty-nine days younger, still shows energy in anything he puts his hands to.”

Mrs. Henry Alphonse of Concord, Mo., over 105, retained her memory and eyesight without glasses till after 104. Mr. Charles Crowley died at Suncook, N. H. over 104. Frank Bogkin, a colored man of Montgomery, Ala., was believed to be 115 at his death recently. When he was about 60 years old, he earned money and purchased his freedom. Tony Morgan, a blind negro, was recently living at Mobile, 105 years old. Pompey Graham of Montgomery, N. Y., lately died at 119, and retained his faculties. Phebe Jenkins of Beaufort County, South Carolina, was believed to be 120 years old when she died about a year ago. Mrs. Louisa Elgin of Seymour, Indiana, whose mother lived to be 115, was recently living at 105.

“Jennie White, a colored woman, died in St. Joseph, Mo., Monday last, aged 122 years. She was born in the eastern part of Georgia, and when twenty years of age was taken to Tennessee, where she remained for ninety-six years. She had lived in St. Joseph about ten years. She was a cook for Captain Waterfall, of George Washington’s staff, during the war of the Revolution. She remembered the death of Washington well, and used to tell a number of interesting stories about early times. She died in full possession of all her mental faculties, but was a cripple and helpless.”

MALES AND FEMALES.—In the first number of the JOURNAL it was stated that although women were from two to six per cent more numerous in population, more males were born by four to sixteen per cent. This was a typographical error; it should have been from four to six per cent, generally

four. The greatest excess of males is in illegitimate births. The reversal of proportions in the progress of life shows that the male mortality is much greater than the female. Hence the more tranquil habits and greater predominance of the moral nature in women increases their longevity, while the greater indulgence of the passions and appetites, the greater muscular and intellectual force among men, are hostile to longevity. Hence the establishment of a true religion, or the application of the "New Education," will greatly increase longevity. It will also be increased by greater care of health in manufacturing establishments, and by diminishing the hours of labor; for exhausting physical labor not only shortens life but predisposes to intemperance. The injurious effect of excessive toil is shown in the shorter lives of the poor, and is enforced by Finlaison's "Report on Friendly Societies to the British Parliament," which says (p. 211) "The practicable difference in the distribution of sickness seems to turn upon the amount of the *expenditure of physical force*. This is no new thing, for in all ages the enervation and decrepitude of the bodily frame has been observed to follow a prodigal waste of the mental or corporeal energies. But it has been nowhere previously established upon recorded experience that the quantum of sickness annually falling to the lot of man is in a direct proportion to the demands upon his muscular power. So it would seem, however."

Philanthropists should therefore unite in limiting the hours of daily labor to ten or less. But more quiet pursuits have greater endurance; women keeping house have no ten hour limit, and the editor of the JOURNAL generally gives more than twelve hours a day to his daily labor.

A NEGRO 135 YEARS OLD.—The St. Louis *Globe Democrat* says: James James, a negro, and citizen of the United States, who resides at Santa Rosa, Mexico, is probably the oldest man on earth. He was born near Dorchester, S. C., in 1752, and while an infant was removed to Medway River, Ga., in the same year that Franklin brought down electricity from the thunder clouds. In 1772 there was quite an immigration into South Carolina, and his master, James James (from whom he takes his name), moved near Charleston, S. C., in company with a number of his neighbors. On June 4, 1776, when 24 years of age, a large British fleet, under Sir Peter Parker,

arrived off Charleston. The citizens had erected a palmetto-wood fort on Sullivan's Island, with twenty-six guns, manned by 500 troops under Col. Moultrie, and on June 28 the British made an attack by land and water, and were compelled to withdraw after a ten-hours' conflict. It was during this fight that Sergeant Jasper distinguished himself by replacing the flag, which had been shot away upon the bastion on a new staff. His master, James James, manned one of the guns in this fight, and Jim, the subject of this sketch, with four other slaves, were employed around the fort as general laborers. Jim followed his master throughout the war, and was with Gen. Moultrie at Port Royal, S. C., Feb. 3, 1779, when Moultrie defeated the combined British forces of Prevost and Campbell. His master was surrendered by Gen. Lincoln at Charleston, S. C., on Feb. 12, 1780, to the British forces, and this ends Jim's military career.

He remembers of the rejoicing in 1792 throughout the country in consequence of Washington's election to the Presidency, he then being 40 years of age. In this year his first master died, aged about 60 years. Jim then became the property of "Marse Henry" (Henry James), owning large estates and about thirty slaves near Charleston. On account of having raised "Marse Henry," Jim was a special favorite with his master, and was allowed to do as he chose. His second master, Henry, died in 1815, about 55 years of age, and Jim, now at 63 years of age, became the property of James James, Henry's second son. In 1833 the railroad from Charleston to Savannah was completed, then the longest railroad in the world, and Jim, with his master, took a trip over the road, and was shown special favors on account of his age, now 81. James James was ten years of age at his father's death, and when he became of age he inherited large estates, slaves, etc., among whom were "old Uncle Jim" and his family. James James in 1855 moved to Texas with all his slaves. He desired that his slaves should be free at his death, and in 1858 moved into Mexico, so that they could be free before his death. James returned to the United States and died in Texas, and in 1865, after there were no longer slaves in the United States, Uncle Jim's children and grandchildren returned to the United States. Five years ago, at the age of 130, Jim could do light chores, but subsisted mostly by contributions from

the citizens, but for the past two years, not being able to walk, he remains for the most part in his little jacal, his wants being supplied by generous neighbors. The rheumatism in his legs prevents him from walking.

So many cases of great longevity have recently been announced, that their detailed publication would be tedious. The New York *Sun* says: "A town in Cuba prides itself upon being the home of eleven women, each of whom is over 100 years of age." According to the census of Germany, December, 1875, there were 160 persons over 100 years of age, of whom there was one woman of 115 years, and another of 117, one man of 118, and another of 120. Our own country has a better record of longevity than this.

Let us rest content with the fact that the world has many centenarians, and that we too are free to live a hundred years, if our ancestors have done their duty in transmitting a good constitution, and we have done our duty in preserving it.

Miscellaneous Intelligence.

AN IMPORTANT DISCOVERY.—In the New Education I have endeavored to show that there are qualities of the atmosphere which science has not yet recognized, which are of the highest importance to human health, and that an atmosphere may have vitalizing or devitalizing qualities with apparently the same chemical composition, because some vitalizing element has been added or subtracted.

This vitalizing element, though analogous to electricity, is not identical with it. We find it absent in a room that has been recently plastered, and is not quite dry. Sleeping in such a room is positively dangerous. We find the same negative depressing condition wherever evaporation has been going on in the absence of sunlight, which appears to supply the needful element.

As evaporation carries off this vitalizing element, precipitation or condensation seems to supply it, especially precipitation from the upper regions of the atmosphere to which it is carried by evaporation, and to which it is supplied by sunshine. Hence we experience a delightful freshness of the atmosphere after a summer shower, or on a frosty morning, when the moisture is not only precipitated, but condensed into frost. Frost gives off more of the exhilarating element of watery vapor than dew, because it is a step farther in condensation. Hence there is a healthful, bracing influence in cold climates, where all the moisture is firmly frozen, and a very unpleasant, depressing influence when a thaw begins. The vicinity of melting snow, or a melting iceberg, is unpleasant and promotive of catarrh and pulmonary diseases.

The pleasant influence of the fresh shower ceases when the fallen moisture begins to evaporate, and the dewy freshness of the early morn

before sunrise ceases as the dew evaporates. The most painfully depressing atmosphere is that which sometimes comes in cold weather from Northern regions which have long been deprived of sunshine.

This element of health, which physiologists have neglected to investigate, has recently been sought by Dr. B. W. Richardson of England. The Popular Science News (of Boston) says:—

“Dr. B. W. Richardson of England, in making some investigations upon the physiological effects of breathing pure oxygen by various animals, has discovered, that, by simply passing the gas a few times through the lungs, it becomes “devitalized,” or incapable of supporting life, although its chemical composition remains the same, and all carbonic dioxide and other impurities are removed. He also found, that, by passing electric sparks through the gas, it became “revitalized,” and regained its usual stimulating effect upon the animal economy. The devitalized oxygen would still support life in cold-blooded animals, and combustible bodies would burn in it as brilliantly as ever. Dr. Richardson considers that, while the gas is in contact with the tissues or blood of a warm-blooded animal, some quality essential to its life-supporting power is lost. The subject is an interesting and important one, and deserves a more thorough investigation.”

JENNIE COLLINS has passed on to her reward above. It would be wrong to neglect mentioning the remarkable career of this devoted woman, who for thirty-five years has been the guardian angel of the poor and struggling women of Boston. Rising from friendless poverty, she became widely known as a champion of human rights, and woman's rights, and, finally, as the founder and indefatigable sustainer of that benevolent institution widely known as Boffin's bower. Her literary powers were finely displayed in a little volume entitled “Nature's Aristocracy,” and her mental vigor was shown in many public addresses. Jennie Collins was a noble illustration of the best form of Spiritualism. She was accompanied, inspired, and sustained

by spirit influence, but did not deem it expedient to let this fact be generally known. The world is not yet enlightened.

GREEK PHILOSOPHY.—The essential pedantic stupidity of Aristotle's logic, and its power to belittle and benumb the intelligence of its reverential students has been shown in every college where this effete study is kept up. We have no better illustration of late than its effect on Prof. Harris, who is a very intelligent and useful citizen, but who has been so befogged by such studies as to suppose that his pedantic talk about syllogisms embodies an important contribution to philosophy, and indeed it was announced as such by his reporter. The superstitious reverence for Greek literature is impressed on all young collegians, and few recover from it. Sir William Hamilton and R. W. Emerson, who were much more intellectual and brilliant than Prof. Harris, were as badly afflicted as he with this Greek superstition, which has been implanted in school boys so young that it dominates their whole lives with the energy of a prenatal condition. The only very silly things ever written by the brilliant Emerson were those passages in which he speaks of Plato; and the silliest thing in the life of Hamilton is the way in which he exulted over some trivial modification of Aristotle's syllogistic ideas, which was about as trivial as that of Prof. Harris, and allowed himself to be publicly flattered by one of his students in the most fulsome manner for the wonderful profundity of his wisdom, that could even add something to the divine wisdom of Aristotle.

To tell a Greek idolater that the divine Plato thought it a great MYSTERY that one and one should make two, that he declared it to be incomprehensible to him, and thought the only possible solution of the mystery to be, that two is produced "*by participation in duality*," would surprise him; but he would be still more surprised to learn that this is only a specimen brick of Plato's divine philosophy, as it abounds in similar puerilities. I have long since reviewed this effete philosophy of an ignorant age, and shown its true character, but my work has never been offered to a bookseller. Yet it shall not be suppressed. The destruction of stultifying

superstitions is as necessary in education and literature as in religion. The ponderous blows of Lord Bacon upon this Greek superstition of the literary classes did not prove fatal, for the same reason that animal organisms of a low, cold-blooded, grade are hard to kill,—they must be cut up in fragments before their death becomes complete; superstitions and beliefs that have no element of intelligent reason, and are perpetuated by social influence, authority, and domination over the young become a blind force that resists all influence from reason.

If my readers are interested in the destruction of venerable and powerful falsehoods that stand in the way of every form of progress, I may be tempted to publish a cheap edition of my work on Greek Philosophy and Logic. It is not in the least presumptuous to lay hands upon this venerable illusion, and show that it has not even the vitality of a ghost. It is but a simulacrum or mirage, and it is but necessary to approach it fearlessly, and walk through it, to discover its essential nonentity.

SYMPOSIUMS deserves a good report. One of the philosophers, whose doctrines were poetically paraphrased in the report of the scientific responses upon human immortality, writes that he enjoyed the poetical paraphrase very much, and never laughed over anything so heartily. It would be pleasant to hear the real sentiments of the remainder. It would be equally interesting to hear how Prof. Harris and the other Concordians enjoy the little sketch of their symposium.

LITERATURE OF THE PAST.—“In an article on the ‘Archetypal Literature for the Future,’ by Dr. J. R. Buchanan, which appears in the JOURNAL OF MAN for March, the writer foreshadows a time to which the American mind is fast advancing when the literature of the past will take its place amongst the mouldering mass which interests the antiquarian, but has no positive influence in guiding the thoughts and actions of the passing generation. There are some indications of a movement in that direction in other countries, though the vast majority, including many Spiritualists and

Theosophists, still explore the records of past ages, looking for the light which is shining all about them in the present, unrealized.”—*Harbinger of Light*, Australia.

THE CONCORD SCHOOL.—We are glad that the Concord School is over, and we should think that the people that have been there would be glad to get home and take part in the things which interest average folks. If people like that sort of thing and can afford it, there is no reason why they should not go there and stay. But to the average man the whole thing looks about as near time wasted as anything which even Boston furnishes to the “uncultured” world outside.—*Boston Record*.

NEW BOOKS.—“THE HIDDEN WAY across the threshold, or the mystery which hath been hidden for ages and from generations,—an explanation of the concealed forces in every man to open the temple of the soul and to have the guidance of the unseen hand.—By J. C. Street, A. B. N., Fellow of S. S. S., and of the Brotherhood Z. Z. R. R. Z. Z.” Lee & Shepard, publishers, Boston (\$3.50). This is a very handsome volume of nearly 600 pages, which I have not had time to examine. It appears to be chiefly a compilation with quotation marks omitted, written in the smooth and pleasing style common in spiritual literature, without any attempt at scientific analysis or criticism. Sharp critics condemn it, but it suits the popular taste and inculcates good moral lessons. I shall examine it hereafter.

“SOLAR BIOLOGY—a scientific method of delineating character, diagnosing disease, determining mental, physical, and business qualifications, conjugal adaptability, etc., etc., from the date of birth.—By HIRAM E. BUTLER, with illustrations.” Boston, Esoteric Publishing Company, 478 Shawmut Avenue (\$5.00). This is a handsome volume, which, from a hasty examination, appears to be a large fragment of Astrology, containing its simplest portion, requiring no abstruse calculations, and hence adapted

to popular circulation. It is meeting with some success, but those who feel much interest in astrology prefer to take in the whole science, which has a much larger number of votaries than is commonly supposed.

DR. FRANZ HARTMANN, of Germany, has published some interesting volumes recently, on "Paracelsus," "White and Black Magic," and "Among the Rosicrucians," which I have had no time to examine. A valuable essay from Dr. Hartmann is on file for publication in the JOURNAL, in which he compares the doctrines of the occult philosophy with those presented in the JOURNAL OF MAN.

PROGRESS OF CHEMISTRY.—FORTY NEW SUBSTANCES.—"During the decade ending with 1886 over forty discoveries of new elementary substances were announced, while the entire number previously known was less than seventy. No less than nine were detected by Crookes last year. The list is likely to be lengthened quite as materially in the current twelvemonth, as A. Pringle already claims to have found six new elements in some silurian rocks in Scotland. Five of these are said to be metals, and the other is a substance resembling selenium, which the discoverer calls hesperisium. One metal is like iron, but does not give some of its reactions; another resembles lead, is quite fusible and volatile, and forms yellow and green salts; another, named erebodium, is black; the fourth is a light-gray powder, and the last is dark in color."

ASTRONOMY.—"The absolute dimensions of a globular star cluster have been studied by Mr. J. E. Gore of the Liverpool Astronomical Society. These clusters consist of thousands of minute stars, possibly moving about a common center of gravity. One of the most remarkable of these objects is 13 Messier, which Proctor thinks is about equal to a first magnitude star. Yet Herschel estimated that it is made up of fourteen thousand stars. The average diameter of each of these components must be forty-five thousand

two hundred and ninety-eight miles, and each star in this wonderful group may be separated from the next by a distance of nine thousand million miles.”

“According to the computations of M. Hermite, a French astronomer, the total number of stars visible to the naked eye of an observer of average visual power does not exceed 6000. The northern hemisphere contains 2478, and the southern hemisphere contains 3307 stars. In order to see this number of stars, the night must be moonless, the sky cloudless, and the atmosphere pure. The power of the naked eye is here stayed. By the aid of an opera glass 20,000 can be seen, and with a small telescope 150,000, while the most powerful telescopes will reveal more than 100,000,000 stars.”

“M. Ligner, an Austrian meteorologist, claims to have ascertained after careful investigation that the moon has an influence on a magnetized needle, varying with its phases and its declination. The phenomenon is said to be more prominently noticeable when the moon is near the earth, and to be very marked when she is passing from the full to her first or second quarter. The disturbances are found to be in their maximum when the moon is in the plane of the equator, and greater during the southern than it is during the northern declination.”

GEOLOGY ILLUSTRATED.—I have often thought that when coal mines are exhausted and land is too valuable to be devoted to raising timber, it may become necessary to draw on the subterranean heat of the earth. This idea is already verified in Hungary.

Late advices say: “The earth’s internal heat is now being used in a practical way at Pesth, where the deepest artesian well in the world is being sunk to supply hot water for public baths and other purposes. A depth of 3120 feet has already been reached, and the well supplies daily 176,000 gallons of water, heated to °150 Fahr.”

A MATHEMATICAL PRODIGY.—Reub Fields, living a few miles south of Higginsville, Mo., though he has no education whatever, and does not know a single figure or a letter of the alphabet, is a mathematical wonder. Though he never carries a watch, he can tell the time to a minute. When asked on what day of the week the 23d of November, 1861 came, he answered, “Saturday.” When asked, “From here to Louisiana, Mo. it is 159 miles; how many revolutions does the driving wheel of an engine fifteen feet in circumference make in a run from this place to Louisiana?” he replied, “55938 revolutions.” Reub was born in Kentucky, and claims that this power was given to him from heaven when he was eight years old, and that the Lord made but one Samson, one Solomon, and one Reub Fields, for strength, wisdom, and mathematics.

ASTROLOGY IN ENGLAND.—Mrs. L. C. Moulton, correspondent of the *Boston Herald*, writes: “In old times a court astrologer used to be kept, as well as a court jester; but I confess I was not aware, until last night, that the astrologer of to-day might be as important to one’s movements as one’s doctor or one’s lawyer. One of the cleverest and busiest literary men in all London said to me last night that he thought the neglect of astrological counsel a great mistake. ‘I have looked into the subject rather deeply,’ he said, ‘and the more I search, the more convincing proof I find of the influence of the stars upon our lives; and now I never begin a new book, or take a journey, or, in short, do anything of any importance without consulting my astrologer.’ And then he went on to tell me the year in which the cholera devastated Naples he had thought of going there. Happily, he consulted his astrologer and was warned against it. In accordance with the astrologer’s advice, he gave up the journey; and just about the time he would otherwise have gone, news came of the cholera visitation. Last year he was warned against a certain journey—told that if he took it he would be ill. For once he defied the stars, and, in consequence, he was taken seriously ill with the very symptoms the astrologer had predicted. But, alas, his astrologer is fat and old—and what shipwreck may not my friend make of

his life when the stars have reclaimed their prophet, and the poor fellow has to struggle on uncounselled!”

PRIMOGENITURE ABOLISHED.—“By a majority of eleven the House of Lords has abolished primogeniture in cases of intestacy. Thus, unless it is formally specified by will, property will henceforth be divided equally among heirs, as in this country. No longer will the eldest son, by the mere fact of the death of his father, come into possession of the estate to the exclusion of his brothers and sisters. Of course, entailed estates will not be affected, and property can be transmitted by will at the testator’s pleasure, but the notable point is that primogeniture cannot henceforth be looked upon as an institution so characteristic and time-honored that departure from it would be a really questionable proceeding.”

MEDICAL INTOLERANCE AND CUNNING.—The proscriptive medical law of Iowa does not seem to be very effective, as it is believed to be unconstitutional, and its friends have been challenged to make test cases, but have not yet begun to enforce it. In Illinois they have a law that is imperative enough against practitioners without diplomas; but as this did not reach those who used no medicines, they have succeeded in procuring a law to reach them also by a new definition of “practicing medicine,” which the new statute says shall include all “who shall treat, operate on, or prescribe for any physical ailment of another.” This would seem sufficient to protect the M. D.’s against all competition, but there is some doubt whether such legislation can be enforced, as it is certainly a corrupt and selfish measure that was never desired by the people. The *Religio Philosophical Journal* speaks out manfully, and “advises all reputable healers of whatever school, to possess their souls in peace, and go steadily forward in their vocation, fearing neither Dr. Rauch nor the unconstitutional provisions of the statutes, under which he and his confederates seek to abridge and restrict the rights of the people. If any reputable practitioner of the healing art, who treats without drugs, is molested in his or her practice,

let them invite prosecution, and communicate with the *Religio Philosophical Journal* for further advice and assistance.” I regret to say there is a strong probability that the friends of medical freedom in Massachusetts will be again called upon to resist attempts to procure medical legislation.

NEGRO TURNING WHITE.—A colored man named Antone Metoyer has been employed at the railroad works in this city (Sacramento) for some time, and his steadiness and industry have caused him to be esteemed by those acquainted with him. Seven or eight months ago his skin was black, but it commenced to turn white, and now his body, arms, legs and neck are as white as those of any Caucasian. The original color is now only upon his face, extending back of the ears, just beneath the chin, and across the upper portion of the forehead, making him appear to be wearing a close-fitting black or dark brown mask. On the chin and nose the dark color is beginning to wear away, and he thinks in a few weeks he will be perfectly white. His hair and whiskers are black and curly. Medical men have taken much interest in his case, and attribute the change in complexion to the effect upon his system of working constantly with potash and other material used in washing greasy waste. He has been advised that it may be dangerous for him to continue under this influence, but he declares that he will stay until the process he is undergoing is completed, if it kills him.—*Record Union*.

THE CURE OF HYDROPHOBIA.—“The English committee appointed by the local government board in April, 1886, to inquire into Pasteur’s inoculation method for rabies, report that it may be deemed certain that M. Pasteur has discovered a method of protection from rabies comparable with that which vaccination affords against infection from smallpox.” As many think there is no protection at all, the question is not finally settled. It is only the stubborn ignorance of the medical profession which gives to Pasteur’s experiments their great celebrity and importance. Other methods have been far more successful than Pasteur’s. Xanthium, Scutellaria (Skull-cap), the

vapor bath, and chloroform or nitrous oxide are more powerful and reliable than any morbid inoculation.

JOHN SWINTON'S paper, at New York, has come to an end. Swinton was a bold, eloquent, and fearless advocate of human rights as he understood them. His failure is an honor to him, and his name will be remembered. Perhaps if he had imitated the Boston dailies, by giving ten to eighteen columns to the record of base ball games, he might have put money in his purse, instead of losing it.

In marked contrast to John Swinton's failure, observe the success of the *New York Tribune*, a newspaper founded by Horace Greeley, but which, since his death, has given, in its unscrupulous course, a good illustration of the Satanic press. The *Boston Herald* says: "The *New York Tribune* is perhaps as good an illustration of the old-fashioned partisan journal as there is in the country. There was an amusing reminiscence of the methods that used to be practised when the *Tribune* was found claiming the Legislature of Kentucky as having been carried by the Republicans in the late elections. The fact was that the Democratic majority in that body was about five to one, and there was really no excuse in a metropolitan journal for not knowing such to be the case." The *Tribune* once complimented highly the JOURNAL OF MAN, but that was when Horace Greeley was alive.

WOMEN'S RIGHTS AND PROGRESS.—The last legislature of Pennsylvania passed a very radical law, providing that marriage shall not impose any disability as to the acquisition or management of any kind of property, making any contracts, or engaging in any business. However, she is not authorized to mortgage her real estate without her husband's co-operation, nor become endorser for another alone. As to making a will she has the same rights as a man.

Ohio has also advanced woman's rights by enabling both husband and wife to dispose of property as if unmarried, and by giving each party one-third life interest in the other's real estate.

In Kansas, women can vote in city and town affairs, and hold municipal and town offices. In one town they have a female mayor. The supreme court of Kansas has decided that when a woman marries she need not take her husband's name unless she chooses.

CO-EDUCATION is successful, nearly every prominent college is beginning to admit women, and they often carry off the prizes from the men. Exclusive masculine colleges will soon rank among the barbarisms of the past.

Female education is advancing in Russia. The universities had 779 female students in 1886, 437 of whom were daughters of noblemen and official personages. On the other hand the Prussian Minister of Education refuses to admit women as regular students at any university or medical school.

Several Italian ladies have distinguished themselves in legal knowledge, and the propriety of their admission to the bar is extensively discussed. About nine-tenths of the newspapers favor their admission.

The practical question, which is most important to the welfare of women, is profitable employment. Miss Simcox says that there are about three millions of women in England engaged in industrial employments, while a large proportion of them, especially in London, have such poor wages as to produce continual suffering. INDUSTRIAL EDUCATION, alike for boys and girls, is the true remedy, worth more than all the nostrums of politicians and demagogues.

SPIRIT WRITING.—Our handsome young friend, Dr. D. J. Stansbury, a graduate of the Eclectic Medical College of New York, is giving astonishing demonstrations on the Pacific coast. When a pair of closed slates is brought, he barely touches them, and the spirit writing begins. Sometimes the slates are held on the head or shoulders of the visitor. At one of his seances at Oakland, it is said that he held the slates for thirty-five persons within two hours, and obtained for each a slate full of writing in answers to questions

placed between the slates. At a public seance in Santa Cruz, following a lecture, folded ballots were sent up by the audience and the answers were sometimes written on closed slates and sometimes by the doctor's hands. Dr. S. has also succeeded in repeating the famous performance of Charles Foster—the names of spirits appearing on his arm in blood-red letters.

PROGRESS OF THE MARVELLOUS.—The *Boston Herald* of Aug. 7 has a long account of the marvellous fires which occur in the house at Woodstock, New Brunswick, of Mr. Reginald C. Hoyt. The people of the town are greatly excited about it, and great crowds gather to witness it, but no one can explain it. The fires break out with no possible cause in the clothes, the carpet, the curtains, bed quilts, or other objects, as much as forty times in a day. The family are greatly worried and alarmed, and have been driven out of the house. The *Herald* reporter went to examine, but found it an entire mystery.

A similar outbreak of fires has been reported in Pennsylvania, at the house of Thomas McKee, a farmer at Turtle Creek. For some weeks the invisible powers have been throwing things about in a topsy turvey way. Since that, flames break out suddenly in the presence of the family, and round holes are burned in the bed-clothes, towels, hats, dresses, and even packages of groceries in the pantry.

PRACTICAL UTILITY OF ANTHROPOLOGY.

(Continued from [page 32.](#))

There is no great reform, no elevation of humanity without understanding MAN,—the laws of his culture, the possibilities within his reach, the extent of the short-comings which exist to-day, the very numerous agencies of brain-building and soul-culture, the wiser methods of the school, the magnetic influences which are sometimes all potent, the dietary, the exercises of body and voice, the power of music and disciplined example, the lofty outreachings for a higher life to which we are introduced by psychometry, the supernal and divine influences which may be brought to bear, and many nameless things which help to make the aggregate omnipotent over young life, but which, alas, are unknown in colleges to-day, and will continue unknown until Anthropology shall have taken its place as the guide of humanity.

P.S.—The doctrine so firmly maintained in this chapter that men are incompetent to judge themselves, and need a scientific monitor of unquestionable authority, has long been recognized. The Catholic confessional is a recognition and application of the principles of great value. But the confessional of the narrow-minded and miseducated priest should be superseded by the confessional and the admonition of Anthropology.

Sterne, in his *Tristram Shandy*, says, “Whenever a man’s conscience does accuse him (as it seldom errs on that side), he is guilty, and unless he is melancholy and hypochondriac, there is always sufficient ground for the accusation. But the converse of the proposition will not hold true,” that if it does not accuse, the man is innocent.

“Thus conscience, placed on high as a judge within us, and intended by our Maker as a just and equitable one too, takes often such imperfect cognizance of what passes, does its office so negligently, often so corruptly, that it is not to be trusted alone, and, therefore, we find there is a necessity, an absolute necessity, of joining another principle with it.”

That “other principle” demanded by Sterne has never been found, until, in the revelation of the functions of soul and brain, we have found the

absolute standard of character, and in Cranioscopy and Psychometry the perfect method of applying the principle to each individual.

An amusing illustration occurred lately in England, which was published as follows:—

“When the address to the queen at the opening of the English royal courts was under consideration by the judges, one very eminent judge of appeal objected to the phrase ‘conscious as we are of our shortcomings.’ ‘I am not conscious of shortcomings,’ he said, ‘and if I were I should not be so foolish as to say so;’ whereupon a learned lord justice blandly observed, ‘Suppose we say, “conscious as we are of each other’s shortcomings.”’”

Chapter VIII—The Origin and Foundation of the New Anthropology

Difficulties of imperfect knowledge in my first studies—First investigation of Phrenology—Errors detected and corrected—The PATHOGNOMIC SYSTEM organized—A brilliant discovery and its results—Discovery of the sense of feeling and development of Psychometry—Its vast importance and numerous applications—The first experiments on the brain and the publication of Anthropology—The discovery of Sarcognomy and its practical value—Reception of the new Sciences—Honorable action of the venerable Caldwell.

The very brief exposition of the structure and functions of the brain already given, may serve as an introduction to the subject and prepare the reader to appreciate the laborious investigations of many years, by means of which so comprehensive a science was brought into existence amid the hostile influences of established opinions and established ignorance.

It is necessary now to present this statement to enable the reader to realize more fully the positive character of the science.

My life has been devoted to the study of man, his destiny and his happiness. Uncontrolled in education, I learned to endure no mental restraint, and, thrown upon my own resources in boyhood, difficulties but strengthened the passion for philosophical knowledge. Yet more formidable difficulties were found in the limited condition of human science, alike in libraries and colleges.

Anthropology, my favorite study, had no systematic development, and the very word was unfamiliar, because there was really nothing to which it could justly be applied. Its elementary sciences were in an undeveloped state, and some of them not yet in existence. Mental philosophy was very limited in its scope, and had little or nothing of a practical and scientific

nature. The soul was not recognised as a subject for science. The body was studied apart from the soul, and the brain, the home of the soul, was enveloped in mystery—so as to leave even physiological science shrouded in darkness, as the central and controlling organ of life was considered an inaccessible mystery. In studying medicine, it seemed that I wandered through a wilderness without a compass and with no cardinal points.

Phrenology promised much, and I examined it cautiously. It struck me at first as an unsatisfactory system of mental philosophy, and I stated my objections before its most celebrated and venerable champion, in public, who assured me that I would be satisfied by further investigation. As it seemed a very interesting department of natural science, I began by comparing the heads of my acquaintances with the phrenological map, and discovering so many striking coincidences that I was gradually satisfied as to its substantial truth, and I do not believe that any one has ever thus tested the discoveries of Gall and Spurzheim, without perceiving their *general* correctness, while many, with less critical observation, have accepted them as absolutely true.

My interest increased with the extent of my observations, until, for several years, I abandoned practical medicine for the exclusive study of the science of the brain in the great volume of nature, with the doctrines of Gall as the basis of the investigation. As it was my purpose to seek the deficiencies as well as the merits of the new science, I tested its accuracy by the careful examination of living heads and skulls in comparison with ascertained character, and with the anatomy of the brain, not forgetting the self-evident principles of mental philosophy. Many thousand critical examinations were made between the years 1834 and 1841, leading to many positive conclusions. The first year's observations made me distinctly aware and certain of several defects in the doctrines, as to the functions ascribed to certain localities of the brain to which were ascribed, Mirthfulness, Acquisitiveness, Adhesiveness, Constructiveness, Tune, Ideality, Combativeness, Destructiveness, and Cautiousness. The functions of these localities were evidently misunderstood, and the faculties erroneously located.

The external senses were omitted from the catalogue of cerebral organs, though evidently entitled to recognition, and the physiological powers of the brain, the prime mover and most important part of the constitution, were almost totally ignored.

Following the old route of exploration by cranioscopy, I sought to supply these defects. I found the supposed Mirthfulness to be a planning and reasoning organ, and the true Mirthfulness to be located more interiorly. Acquisitiveness was evidently located farther back. The so-called organ of Adhesiveness appeared to be incapable of manifesting true friendship, and its absence was frequently accompanied by strong capacities for friendship, of a disinterested character. Constructiveness appeared to be located too low, and too far back, running into the middle lobe, which is not the place for intellect. Tune did not appear to correspond regularly to musical talent. Many of the higher functions ascribed to Ideality were conspicuous in heads which had that organ small, with a large development just above it. Combaticiveness had evidently less influence upon physical courage than was supposed, for it was sometimes well developed in cowards, and rather small in brave men. Cautiousness was evidently not the organ of fear, for the bravest men, of whom I met many in the southwest, sometimes had it in predominant development, and in the timid it was sometimes moderate, or small. Destructiveness was frequently a characteristic of narrow heads (indeed this is the case with the Thugs of India), and a broad development above the ears was sometimes accompanied by a mild disposition. The height of the head above the ears did not prove a correct criterion of moral character, nor did the breadth indicate correctly the amount of the selfish and violent passions.

I observed that the violent and selfish elements of character were connected with occipital depth, and elongation; that the affections were connected with the coronal region, that the sense of vision was located in the brow, and the sense of feeling in the temples, near the cheekbone, that the upper occipital region was the seat of energetic powers, and the lower, of violent or criminal impulses, and that the whole cerebrum was an apparatus of mingling convolutions, in which the functions, gradually

changing from point to point, presented throughout a beautiful blending and connection.

Observing daily the comparative development of brain and body, with their reciprocal influence, I traced the outlines of cerebral physiology, and the laws of sympathetic connection or correspondence between the body and the encephalon, by which, in a given constitution, I would determine from the head the development of the whole body, the peculiar distributions of the circulation, with the consequent morbid tendencies, the relative perfection of the different senses and different organs of the body, and the character of the temperament.

Seeking continually for the fundamental laws of Anthropology, criticising and rejecting all that appeared objectionable or inconsistent, I acquired possession of numerous sound and comprehensive principles concerning the fundamental laws of cerebral science, which were at once touchstones for truth and efficient instruments for further research.

These fundamental laws, though very obvious and easily perceived when pointed out, had been overlooked by my predecessors, but are always accepted readily by my auditors, when fully explained. As new facts and principles led to the discovery of other facts and principles, a system of philosophy (not speculative, but scientific) was thus evolved, and a number of geometrical principles were established as the basis of the science of the brain, so evidently true, though so long overlooked, as to command the unanimous assent of all to whom they have been presented; and, as the acceptance of these principles involves the general acceptance of cerebral science, my labors as a teacher have ever been singularly harmonious, and free from doubt, antagonism, and contention.

The fundamental principle of the philosophy was geometric or mathematical, as it examined the construction of the brain, and showed an exact mathematical relation between each organ of the brain and its effects on the body, in the spontaneous gestures, the circulation of blood, the nervous forces, and local functions. Its leading characteristic being the law of the expression of the vital forces and feelings in outward acts. This doctrine was called the PATHOGNOMIC SYSTEM.

I was preparing to publish in several volumes the reorganized science as the Pathognomic System, when the consummation of my researches, by a brilliant discovery, led me into a new world of knowledge—to the full development of the science of Anthropology, according to which the brain gives organic expression to functions which are essentially located in the soul, and the body gives organic manifestation to functions which are controlled in the brain, while the body reacts upon the brain and the brain upon the soul. Thus, every element of humanity has a triple representation—that in the soul, which is purely psychic, yet by its influence becomes physiological in the body; that in the body which is purely physiological, yet by its influence becomes psychic in the soul, and that in the brain which produces physiological effects in the body, and psychic effects in the soul.

Thus, each of the three repositories of power is a psycho-physiological representation of the man; more physical in the body, more spiritual in the soul, but in the brain a more perfect psycho-physiological representation of man as he is in the present life. This full conception of the brain, which Gall did not attain, involved the new science of CEREBRAL PHYSIOLOGY, in which the brain may express the character of the body, as well as the soul, of which I would only say at present that my first observations were directed to ascertaining the cerebral seats of the external senses, vision, hearing and feeling, and the influences of different portions of the brain on different portions of the body.

The location of the sense of feeling, of which I became absolutely certain in 1838, at the base of the middle lobe has since been substantially confirmed by Ferrier's experiment on the monkey; but I have not been concerned about the results of vivisection, knowing that if I have made a true discovery, vivisection and pathology must necessarily confirm it; and I am not aware that any of my discoveries have been disturbed by the immense labors of vivisection.

The discovery of the organ of the sense of feeling led to an investigation of its powers, and the phenomena exhibited when its development was unusually large—hence came the initial fact of psychometry. Early in 1841 I found a very large development of the organ, in the head of the late

Bishop Polk, then at Little Rock, the capital of Arkansas, who subsequently became a confederate general. After explaining to him his great sensibility to atmospheric, electric, and all other physical conditions, he mentioned a still more remarkable sensibility—that whenever he touched brass, he had immediately the taste of brass in his mouth, whether he knew what he was touching or not. I lost no time in verifying this observation by many experiments upon other persons, and finding that there were many in whom sensibility was developed to this extent, so that when I placed pieces of metal in their hands, behind their backs, they could tell what the metal was by its taste, or some other impression. Further examinations showed that substances of any kind, held in the hands of sensitives, yielded not only an impression upon the sense of taste, by which they might be recognized, but an impression upon the entire sensibility of the body. Medicines tried in this manner gave a distinct impression—as distinct as if they had been swallowed—to a majority of the members of a large medical class, in the leading medical school at Cincinnati, and to those who had superior psychometric capacities, the impression given in this manner enabled them to describe the qualities and effects of the medicines as fully and accurately as they are given in the works on *materia medica*.

This method of investigation I consider not only vastly more easy and rapid than the method adopted by the followers of Hahnemann, but more accurate and efficient than any other method known to the medical profession, and destined, therefore, to produce a greater improvement in our knowledge of the *materia medica* than we can derive from all other methods combined, in the same length of time. I may hereafter publish the practical demonstration of this, but the vast amount of labor involved in my experimental researches has not yet permitted me to take up this department, although it has yielded me some very valuable discoveries.

It may require a century for mankind fully to realize the value of Psychometry. It has been clearly, though I cannot say completely shown in the “*MANUAL OF PSYCHOMETRY*,” to which I would refer the reader. I would simply state that the scientific discovery and exposition of Psychometry is equivalent to the dawn of new intellectual civilization, since it enables us to

advance rapidly toward perfection all sciences and forms of knowledge now known, and to introduce new sciences heretofore unknown.

1.

To the MEDICAL COLLEGE it will give a method of accurate diagnosis which will supersede the blundering methods now existing—a method of RAPIDLY enlarging and perfecting the materia medica—a method of exploring all difficult questions in Biology and Pathology, and a complete view of the constitution of man.

2.

To the UNIVERSITY it offers a method of revising and correcting history and biography—of enlarging our knowledge of Natural History, Geology, and Astronomy, and exploring Ethnology.

3.

To the CHURCH it offers a method of exploring the origins of all religions, the future life of man, and the relations of terrestrial and celestial life.

4.

To the PHILANTHROPIST it offers the methods of investigating and supervising education and social organization which may abolish all existing evils.

The foregoing were the initial steps and results in the development of Psychometry, simultaneously accompanied by those other discoveries in 1841, the scope and magnitude of which appear to me and to those who have studied my demonstrations, to be far more important than anything that has ever been discovered or done in Biological science, being nothing less than a complete scientific demonstration of the functions of the brain in all its psycho-physiological relations. To appreciate their transcendent importance, it is necessary only to know that the experiments have been carefully made, have often been repeated during the past forty-five years, and that all they demonstrate may also be demonstrated by other means, and fully established, if no such experiments could be made.

The origin of this discovery was as follows. My advanced investigations of the brain, between 1835 and 1841, had added so much to the incomplete and inaccurate discoveries of Gall, and had brought cerebral science into so much closer and more accurate relation with cerebral anatomy and embryology, as illustrated by Tiedemann, that I became profoundly aware of the position in which I found myself, as an explorer, possessed of knowledge previously quite unknown, and yet, at the same time, however true, not strictly demonstrable, since none could fully realize its truth without following the same path and studying with the same concentrated devotion the comparative development of the brain in men and animals. Such zeal, success, and assiduity I did not believe could be expected. There might not be one man in a century to undertake such a task (for all the centuries of civilization had produced but one such man—the illustrious Gall), and when he appeared his voice would not be decisive. I would, therefore, appear not as presenting positive knowledge, but as contributing another theory, which the medical profession, regardless of my labors, would treat as a mere hypothesis.¹

It was absolutely necessary that the functions of the brain should be demonstrated as positively as those of the spinal nerves had been demonstrated by Majendie and Bell. Two methods appeared possible. The two agents were galvanism and the aura of the nervous system, commonly called animal magnetism. My first experiments in 1841, satisfied me that both were available, but that the *nervaura* was far more available, efficient, and satisfactory. Upon this I have relied ever since, though I sometimes experiment with galvanism, to demonstrate its efficiency, and Dr. De la Rua, of Cuba, informed me over twenty years ago that he found very delicate galvanic currents available for this purpose in his practice.

Animal magnetism or mesmerism had been involved in mystery and empiricism. There had never been any scientific or anatomical explanation of the phenomena, and this mystery I desired to dispel. My first step was to ascertain that for experiments on the nervous system we did not need the somnambulic or hypnotic condition, and that it was especially to be avoided as a source of confusion and error. Whenever the organ of sensibility, or

sensitiveness, was sufficiently developed and predominant, the conditions of neurological experiments for scientific purposes were satisfactory, and to make such experiments, the subjects, instead of being ignorant, passive, emotional, hysteric, or inclined to trance, should be as intelligent as possible, well-balanced and clear-headed,—competent to observe subjective phenomena in a critical manner. Hence, my experiments, which have been made upon all sorts of persons, were most decisive and satisfactory to myself when made upon well-educated physicians, upon medical professors, my learned colleagues, upon eminent lawyers or divines, upon strong-minded farmers or hunters, entirely unacquainted with such subjects, and incapable of psychological delusion, or upon persons of very skeptical minds who would not admit anything until the phenomena were made very plain and unquestionable.

While the nervaura of the human constitution (which is as distinctly perceptible to the sensitive as its caloric and electricity) is emitted from every portion of the surface of the head and body, the quality and quantity of that which is emitted from the inner surface of the hand, render it most available, and the application of the hand of any one who has a respectable amount of vital and mental energy, will produce a distinct local stimulation of functions wherever it may be applied upon the head or body. In this manner it is easy to demonstrate the amiable and pleasing influence of the superior regions of the brain, the more energetic and vitalizing influence of its posterior half, and the mild, subduing influence of the front.

In my first experiments, in the spring of 1841, I found so great susceptibility that I could demonstrate promptly even the smallest organs of the brain, and it was gratifying to find that the illustrious Gall had ascertained, with so marvellous accuracy the functions of the smallest organs in the front lobe, and the subject could be engrossed in the thought of numbers and counting by touching the organ of number or calculation. Eagerly did I proceed in testing the accuracy of all the discoveries of Gall and the additions I had made by craniological studies, as well as bringing out new functions which I had not been able to anticipate or discover. Omitting the history of those experiments, I would but briefly state that in

1842 I published a complete map of the brain, in which the full development of human faculties made a complete picture of the psycho-physiological constitution of man, and thus presented for the first time a science which might justly be called *Anthropology*.²

It is obvious that prior to 1842 there was nothing entitled to the name of ANTHROPOLOGY, as there was no complete geography before the discovery of America and circumnavigation of the globe. When man is fully portrayed by the statement of all the psychic and all the physiological faculties and functions found in his brain, which contains the totality, and manifests them in the soul and body, it is obvious that we have a true Anthropology, which, to complete its fulness, requires only the study of the soul as an entity distinct from the brain, and of the body as an anatomical and physiological apparatus. The latter had already been well accomplished by the medical profession, and the former very imperfectly by spiritual psychologists. But neither the physiology, nor the pneumatology had been placed in organic connection with the central cerebral science.

In consummating such tasks, I felt justified, in 1842, in adopting the word Anthropology, as the representative of the new science, though at that time it was so unfamiliar as to be misunderstood. This science, as presented in my *Outlines of Anthropology* in 1854, embraced another very important and entirely novel discovery—the psycho-physiological relations of the surface of the body, the manner in which every portion of the body responds to the brain and the soul, the final solution of the great and hitherto impenetrable mystery of the triune relations of soul, brain, and body. This discovery, constituting the science of Sarcognomy, became the basis of a new medical philosophy, explaining the influence of the body on the soul, in health, and disease, and the reciprocal influence of the soul on the body.

This manifestly modified our views of therapeutics and revolutionized electro-therapeutics by pointing out the exact physiological and psychic effects of every portion of the surface of the body, when subject to local treatment, and hence, originating new methods of electric practice, in which many results were produced not heretofore deemed possible. All this was

fully presented in my work on THERAPEUTIC SARCOGNOMY, published in 1885, which was speedily sold.

In contemplating these immense results of a successful investigation of the functions of the brain, I can see no logical escape from the conclusion that such a revelation of the functions of the brain is by far the most important event that belongs to the history of vital science—an event so romantically different from the common, slow progress of science when cultivated by men of ability, that I do not wonder at the incredulity which naturally opposes its recognition, and seems to render the most unanimous and conclusive testimony from honorable scientists apparently ineffective. The support of the medical college in which I was Dean of the Faculty, the hearty endorsement by the Faculty of Indiana State University, and by numerous committees of investigation, seem to count as nothing with the conservative portion of the medical profession, who have ever understood how to ignore so simple and positive a demonstration as that of Harvey, or so practical a demonstration as that of Hahnemann, or so irresistible a mass of facts as those of modern psychic science.

The question will naturally arise among the enlightened lovers of truth, why so grand and so *demonstrable* a science should for forty-five years have made so little progress toward general recognition. It is sufficient to say that new and revolutionary truth is never welcomed, and, if the discoverer is not active as a propagandist it has no diffusion. I did not feel that there was any receptiveness across the ocean for what was resisted here. Nevertheless I did prepare and send to Edinburgh, in 1841, a brief report of my discoveries accompanied by an endorsement or introduction from the venerable Prof. Caldwell, the founder of the successful medical college at Louisville, whose lectures were attended by four hundred pupils. I supposed the gentlemen of the Phrenological Society at Edinburgh the most liberal parties in Great Britain, but they declined publishing my memoir as *too marvellous*, and proposed merely to file it away as a caveat of the discovery. That ended all thoughts of Europe; and, indeed, it seemed to me premature to urge such a discovery and so grand a philosophy upon the world in the present state of its intellectual civilization. I ceased to

agitate the subject for many years, and allowed myself to be drawn into the political agitations connected with our civil war, to mitigate some of its social and political evils.

Of late, however, an urgent and imperative sense of duty has put my pen in motion as the remnant of my life will be hardly sufficient to record the results of my investigations.

In the “New Education” and the “Manual of Psychometry—the dawn of a new civilization”—I have appealed to the public, and three editions of the former with two of the latter show that the public is not indifferent. The recognition of the marvellous claims of Psychometry will prepare the way for the supreme science of Anthropology, to which the coming century will do justice.

In justice to the learned Prof. Caldwell and myself, I should not omit to mention that this distinguished, eloquent, and venerable gentleman, who, in his early life, was a cotemporary of the famous Dr. Rush, of Philadelphia, and throughout his life was a champion of the most progressive doctrines in Biology, not only gave his friendly co-operation on the first presentation of my discoveries, but ten years later honored me with a visit at Cincinnati, to become more fully acquainted with them, and subsequently, by appointment of the National Medical Association, prepared a report upon subjects of a kindred nature, in which he incorporated a statement of my discoveries. His subsequent illness and death, in 1854, at an advanced age, prevented the delivery of this memoir.

In signal contrast to the honorable and candid course of Prof. CALDWELL, and to the candid examination, followed by eulogistic language of Prof. H. P. GATCHELL, ROBERT DALE OWEN, President ANDREW WYLIE, Rev. JOHN PIERPONT, Dr. SAMUEL FORRY, Prof. WM. DENTON, the eloquent Judge ROWAN, and a score of other eminently intellectual men, it is my duty to record the melancholy fact that the great majority of professional men, when tested, have manifested an entire apathy, if not a positive aversion, to the investigations and discoveries in which these momentous results have been reached.

While no aversion, disrespect, or suspicion was shown toward myself, a stubborn aversion was shown to investigations that might have revolutionary results—proving that our false systems of education teach men not to think independently, but to adhere closely to precedent authority, fashion, popularity, and *habit*, which is the inertia of the mental world.

The faculty of my alma mater (excepting Prof. Caldwell) refused to investigate the subject, even when invited by their Board of Trustees. The Boston Academy of Arts and Sciences, embracing the men at the head of the medical profession, pretended to take up the subject, but in a few hours dropped it, with polite compliments to myself, in 1842. The American Medical Association, in 1878, refused to entertain the subject because I could not coincide with them in my sentiments, and accept their code of bigotry. There was no formal action of the Association, but my friend, Prof. Gross, then recognized as the Nestor of the profession, and holding the highest position of authority, informed me semi-officially, very courteously, that none of my discoveries could ever be brought to the notice of the Association, because I did not accept their code. Thus (without mentioning other instances), I have stood before the public with a *demonstrable* science, challenging investigation by critical opponents, who have so uniformly evaded or shrunk from the test that I have ceased to care for their opinions, while I still entertain as profound a respect as ever for the investigations of the candid and manly, among whom I never fail to find friendship and cordiality.

Looking back forty-five years, I remember with extreme pleasure the friendly co-operation of ROWAN and CALDWELL. The American medical profession never had a more dignified, imposing, and high-toned representative than Prof. Caldwell. Nor was the legal profession anywhere ever adorned by a more commanding and gracious representative than the unsurpassed advocate, ROWAN, who was widely known as the “OLD MONARCH.” The nobility of such men was shown in

their noble bearing toward a dawning science, In which they saw the
grandeur of the future.

FOOTNOTES

1. I would mention that in the progress of my discoveries, especially in 1838-39, I came into frequent and intimate association with the late Prof. Wm. Byrd Powell, M. D., the most brilliant, and original of all American students of the brain, whose lectures always excited a profound interest in his hearers, and, in comparing notes with him, I found my own original observations well sustained by his. Though erratic in some of his theories, he was a bold student of nature, and the accidental destruction of his manuscript by fire, when too late in his life to repair the loss, was a destruction of much that would have been deeply interesting.

[Return](#)

2. I do not publish or circulate this map apart from the explanatory volume (Outlines of Anthropology) for the reason that it is impossible by any nomenclature of organs to convey a correct idea of the functions, and hence, such a map would tend to a great many misconceptions. [Return](#)

BUSINESS DEPARTMENT.

COLLEGE OF THERAPEUTICS.

Next Session Begins November 1, 1887.

This institution is the germ of what will be an immense revolution in education hereafter, when the knowledge now given to small classes will hold a conspicuous place in every college, and will be presented in every high school.

The mountain mass of inertia, which opposes, passively, all fundamental changes, cannot now resist scientific demonstration as it has in the past. The instruction in the College of Therapeutics, is thoroughly demonstrative, leaving no room for doubt, and it gives a species of knowledge which ought to be a part of every one's education—a knowledge of the constitution of man, not obtainable to-day in any medical or literary college, nor in our mammoth libraries. It is not merely as a deep philosophy that this interests us, but as a guide in the preservation of health, and in the regulation of spiritual phenomena, which would, to a very great extent, supersede our reliance on the medical profession by giving us the control of the vital powers, by which we may protect ourselves, and control the development of the young.

Each student was made to feel the effects of local treatment on the body, and the power of rapidly changing disease to health, and was personally taught to perform the manipulations for this purpose, and to investigate disease or portray character by the psychometric methods as well as to test the value of medicines.

The various uses and scientific application of electricity were shown, and many things entirely unknown and unrecognized in works on Electro-Therapeutics. The entire class was placed under a medical influence simultaneously by the agency of electricity—an operation so marvelous that it would be considered incredible in medical colleges. By these and other experiments and numerous illustrations and

lucid explanations of the brain and nervous system, the instruction was made deeply interesting, and students have attended more than one course to perfect themselves in the science. The following declaration of sentiments shows how the course was regarded by the class:

“The summer class of 1887 in the College of Therapeutics, feeling it their duty to add their testimony to that of many others in reference to the grand scientific discoveries which they have seen thoroughly demonstrated by Prof. J. R. Buchanan, would say to the public that no one can attend such a course of instruction as we have recently been engaged in, without realizing that Therapeutic Sarcognomy greatly enlarges the practical resources of the healing art for the medical practitioner, magnetizer and electro-therapist, while Psychometry, whose positive truths we have tested and proven, like the sun’s rays, illumines all the dark problems of medical practice and of psycho-physiological sciences.

“Therapeutic Sarcognomy explains the very intricate and mysterious relations of the soul, the brain and body, which prior to Prof. Buchanan’s discoveries were unknown to all scientific teachers, and are even now only known to his students and the readers of his works,

“We feel that we have been very fortunate in finding so valuable a source of knowledge, whose future benefits to the human race, in many ways, cannot be briefly stated, and we would assure all who may attend this college, or read the published works of Prof. Buchanan, and his monthly, the *Journal of Man*, that they will, when acquainted with the subject, be ready to unite with us in appreciating and honoring the greatest addition ever made to biological and psychological sciences. Hoping that the time is not far distant when all students in medical colleges may obtain access to this most important knowledge, we give our testimony to the public.”

H. C. ALDRICH, M. D., D. D. S., *Chairman*.

DR. JNO. C. SCHLARBAUM, *Secretary*.

Enlargement of the Journal.

If the readers of the JOURNAL knew how much very interesting matter is crowded out of each number of the JOURNAL, they would be very anxious for its enlargement.

Advertising in the Journal.

The financial success of monthly magazines, depends much upon a liberal advertising patronage. I would say just to all my readers, that the JOURNAL has a larger circulation than many medical journals which are filled with advertisements. It is an excellent medium for those who have new and valuable things to present, for it circulates among the most progressive and enlightened class of people. The terms are the same which are common in magazines.

☞ An advertising agent might find profitable employment by applying to the editor of the JOURNAL.

Works of Prof. J. R. Buchanan.

THE NEW EDUCATION.—\$1.50.

“It is incomparably the best work on education that I have ever seen.”—Prof. Wm. Denton. “I regard it as by far the best work on education ever published”.—Rev. B. F. Barrett.

MANUAL OF PSYCHOMETRY.—The dawn of a new civilization,—\$2.16.

“The like of this work is not to be found in the whole literature of the past.”—*New York Home Journal*. “He has boldly navigated unknown seas till he has found a far greater and more important world than the Genoese navigator discovered.”—*Hartford Times*. “There are striking reflections upon almost every page, and a richness of language and freshness of spirit that is peculiarly marked.” *Medical Brief*, St. Louis. “A century in advance of his time.”—*People's Health Journal*, Chicago.

PSYCHO-PHYSIOLOGICAL CHART OF SARCOGNOMY.—21×31 inches, \$1. Showing the vital powers of soul, brain, and body in their location, as a guide for treatment. “Upon the psychic functions of the brain, Prof. Buchanan is the highest living authority.”—*American Homœopathist*.

THERAPEUTIC SARCOGNOMY.—Now in preparation, to be published next winter.

OUTLINES OF ANTHROPOLOGY.—Now in preparation.

PRACTICE OF PSYCHOMETRY.—Mrs. C. H. Buchanan continues the practice of Psychometry, 6 James Street, Boston. Personal interview, \$2. Written descriptions, \$3. Elaborate descriptions, \$5. The objects of Psychometry are the description of character, constitution, health, or disease, and such advice as circumstances require.

UNLIKE ANY OTHER PAPER.

The *Spectator*, unlike other home papers, seeks (1) to acquaint every family with simple and efficient treatment for the various common diseases, to, in a word, educate the people so they can avoid disease and cure sickness, thus saving enormous doctors' bills, and many precious lives. (2) To elevate and cultivate the moral nature, awakening the conscience, and developing the noblest attributes of manhood. (3) To give instructive and entertaining food to literary taste, thus developing the mind. (4) To give just such hints to housekeepers that they need to tell how to prepare delicious dishes, to beautify homes, and to make the fireside the most attractive spot in the world.—*Am. Spectator*.

MAYO'S ANÆSTHETIC.

The suspension of pain, under dangerous surgical operations, is the greatest triumph of Therapeutic Science in the present century. It came first by mesmeric hypnotism, which was applicable only to a few, and was restricted by the jealous hostility of the old medical profession. Then came the nitrous oxide, introduced by Dr. Wells, of Hartford, and promptly discountenanced by the enlightened (?) medical profession of Boston, and set aside for the next candidate, ether, discovered in the United States also, but far inferior to the nitrous oxide as a safe and pleasant agent. This was largely superseded by chloroform, discovered much earlier by Liebig and others, but introduced as an anæsthetic in 1847, by Prof. Simpson. This proved to be the most powerful and dangerous of all. Thus the whole policy of the medical profession was to discourage the safe, and

encourage the more dangerous agents. The magnetic sleep, the most perfect of all anæsthetic agents, was expelled from the realm of college authority; ether was substituted for nitrous oxide, and chloroform preferred to ether, until frequent deaths gave warning.

Nitrous oxide, much the safest of the three, has not been the favorite, but has held its ground, especially with dentists. But even nitrous oxide is not perfect. It is not equal to the magnetic sleep, when the latter is practicable, but fortunately it is applicable to all. To perfect the nitrous oxide, making it universally safe and pleasant, Dr. U. K. Mayo, of Boston, has combined it with certain harmless vegetable nervines, which appear to control the fatal tendency which belongs to all anæsthetics when carried too far. The success of Dr. Mayo, in perfecting our best anæsthetic, is amply attested by those who have used it. Dr. Thorndike, than whom, Boston had no better surgeon, pronounced it “the safest the world has yet seen.” It has been administered to children and to patients in extreme debility. Drs. Frizzell and Williams, say they have given it “repeatedly in heart disease, severe lung diseases, Bright’s disease, etc., where the patients were so feeble as to require assistance in walking, many of them under medical treatment, and the results have been all that we could ask—no irritation, suffocation, nor depression. We heartily commend it to all as the anæsthetic of the age.” Dr. Morrill, of Boston, administered Mayo’s anæsthetic to his wife with delightful results when “her lungs were so badly disorganized, that the administration of ether or gas would be entirely unsafe.” The reputation of this anæsthetic is now well established; in fact, it is not only safe and harmless, but has great medical virtue for daily use in many diseases, and is coming into use for such purposes. In a paper before the Georgia State Dental Society, Dr. E. Parsons testified strongly to its superiority. “The nitrous oxide, (says Dr. P.) causes the patient when fully under its influence to have very like the appearance of a corpse,” but under this new anæsthetic “the patient appears like one in a natural sleep.” The language of the press, generally has been highly commendatory, and if Dr. Mayo had occupied so conspicuous a rank as Prof. Simpson, of Edinburgh, his new anæsthetic would have been adopted at once in every college of America and Europe.

Mayo's Vegetable Anæsthetic.

A perfectly safe and pleasant substitute for chloroform, ether, nitrous oxide gas, and all other anæsthetics. Discovered by Dr. U. K. Mayo, April, 1883, and since administered by him and others in over 300,000 cases successfully. The youngest child, the most sensitive lady, and those having heart disease, and lung complaint, inhale this vapor with impunity. It stimulates the circulation of the blood and builds up the tissues. Indorsed by the highest authority in the professions, recommended in midwifery and all cases of nervous prostration. Physicians, surgeons, dentists and private families supplied with this vapor, liquefied, in cylinders of various capacities. It should be administered the same as Nitrous Oxide, but it does not produce headache and nausea as that sometimes does. For further information pamphlets, testimonials, etc., apply to

DR. U. K. MAYO, Dentist,

378 Tremont St., Boston, Mass.

Religio-Philosophical Journal.

ESTABLISHED 1865.

PUBLISHED WEEKLY AT

92 La Salle Street, Chicago,

By JOHN C. BUNDY,

TERMS OF SUBSCRIPTION IN ADVANCE:

One copy, one year \$2.50

Single copies, 5 cents. Specimen copy free.

All letters and communications should be addressed, and
all remittances made payable to

JOHN C. BUNDY, Chicago, Ill.

A Paper for all who Sincerely and Intelligently Seek Truth
without regard to Sect or Party.

Press, Pulpit, and People Proclaim its Merits.

Concurrent Commendations from Widely Opposite Sources.

Is the ablest Spiritualist paper in America.... Mr. Bundy
has earned the respect of all lovers of the truth, by his
sincerity and courage.—*Boston Evening Transcript.*

I have a most thorough respect for the JOURNAL, and
believe its editor and proprietor is disposed to treat the
whole subject of spiritualism fairly.—*Rev. M. J. Savage*
(Unitarian) Boston.

I wish you the fullest success in your courageous course.
—*R. Heber Newton, D. D.*

Your course has made spiritualism respected by the
secular press as it never has been before, and compelled an
honorable recognition.—*Hudson Tuttle, Author and*
Lecturer.

I read your paper every week with great interest.—*H. W.*
Thomas, D. D., Chicago.

I congratulate you on the management of the paper.... I
indorse your position as to the investigation of the
phenomena.—*Samuel Watson, D. D., Memphis, Tenn.*

FACTS,

A MONTHLY MAGAZINE,

DEVOTED TO

Mental and Spiritual Phenomena,

INCLUDING

Dreams, Mesmerism, Psychometry, Clairvoyance,
Clairaudience, Inspiration, Trance, and Physical
Mediumship; Prayer, Mind, and Magnetic Healing; and all
classes of Psychical Effects.

Single Copies, 10 Cents; \$1.00 per year.

PUBLISHED BY

Facts Publishing Company,

(Drawer 5323,) BOSTON, MASS.

L. L. WHITLOCK, Editor.

For Sale by COLBY & RICH, 9 Bosworth Street.

Transcriber's Note: The Table of Contents was copied from the index to the volume. The article on [ANTHROPOLOGY](#) is continued from the [previous issue's page 32](#).

*** END OF THE PROJECT GUTENBERG EBOOK BUCHANAN'S
JOURNAL OF MAN, SEPTEMBER 1887 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the

electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE

LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability,

costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other

ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.